

CS & AI • EXPLAINED

How does a computer store a photo?

To a computer, your photo is just a grid of numbers.

THE HOOK

A photo looks **smooth**, real, full of colour.

But a computer doesn't see a "picture" at all — it sees a giant **spreadsheet of numbers**.

Let's zoom in until you can see them. 🔍

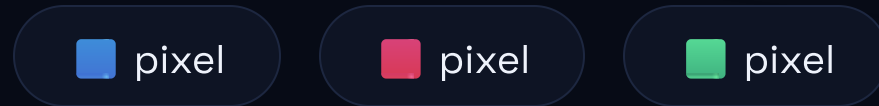


TODAY'S QUESTION

Your selfie is colour and light.
So how does a machine that
only knows **numbers** hold it?

A photo is a grid of **tiny squares**.

- Each little coloured square is a **pixel**.
- Zoom in far enough and you literally see the squares.
- A phone photo has **millions** of them.



Megapixels = millions of pixels. mega just means **million**.

Every colour is just **3 numbers**.

• • •

```
# one pixel = Red, Green, Blue (0-255)
( 255, 0, 0 )    → red
( 0, 0, 0 )      → black
( 255, 255, 255 ) → white
```

- Three lights: **Red**, **Green**, **Blue**.
- Each one a number from **0 to 255** — off to full brightness.
- Mix them and you get **any colour** you want.

IDEA 3 / 4 — WHY PHOTOS ARE "HEAVY"

So a photo = a **huge list** of numbers.

• • •

one photo

millions of pixels × 3 numbers each = a giant list

stored as **binary** # the same 0s and 1s, callback to EP05

- Every number is finally written as **0s and 1s**.
- More pixels → more numbers → a **bigger** file.
- That's why photos are measured in **MB**.

Compression throws away what your **eye** won't miss.

- Storing every single number is **wasteful**.
- **JPEG** drops tiny details you'd never notice...
- ...and reuses **repeated patterns** instead of copying them.
- Result: far smaller than the raw image.

PNG VS JPEG

PNG keeps everything *(lossless)*.

JPEG drops detail *(lossy)*.

WHY IT MATTERS

This explains your whole gallery.

- **Megapixels** = how many pixels the camera captures.
- File **size** = how many numbers had to be stored.
- Zoom in too far or open an old, tiny photo → you see the squares: it looks **blocky / pixelated**.
- Sharing a logo? Pick **PNG**. A normal photo? **JPEG** saves space.

COMMON MYTH — "YEH GALAT SAMAJHTE HAIN"

X MYTH

"The computer stores the actual colours — the real picture is inside the file."

✓ REALITY

It stores **numbers** — the RGB values of each pixel. Your screen turns those numbers back into light. The "photo" only exists when it's **displayed**.

RECAP — 2 THINGS TO REMEMBER

- 1 A photo = a grid of **pixels**, each just a few numbers (**RGB**).
- 2 Compression shrinks the file by cleverly **dropping detail** you won't notice.

Your photo = a grid of numbers.

▶ Subscribe @KatixoKhojLab

Next question: What is "the cloud" actually? 